

Prior-Art Search Report

Search mode: Novelty Subject: — Date: 2026-07-18

Search query / invention

A cordless handheld vacuum lifter for glass and stone panels, with a flexible sealing lip, an electric vacuum pump that keeps running to hold grip on rough or porous surfaces, and a pressure sensor th

Method & corpus

Semantic + agentic search over a 107,795-publication vacuum-gripping corpus (US/EP/WO/DE, all dates; 1,819,616 embedded passages). Retrieval: 8-channel adaptive cascade (dense + BM25 + CPC + citation/family + query-by-example + cross-lingual) with weighted reciprocal-rank fusion and a cross-encoder reranker. Prior-art dating by a jurisdiction-neutral novelty/inventive-step engine.

1/10	5	4	DE, EN
elements disclosed	references cited	novel (uncovered)	languages

Executive summary

Strongest references:

- **US-1294103-A** — Vacuum lifting device. (match 0.779; not dated)
- **US-1178147-A** — Vacuum lifting device. (match 0.779; not dated)
- **US-762499-A** — Vacuum-lifter. (match 0.773; not dated)

Elements disclosed by the cited art (1/10): Vacuum lifter for stone panels

Apparently novel (not found in corpus): Grip on porous surfaces, Grip on rough surfaces, Pressure sensor alarms on grip vacuum loss, Pressure sensor for vacuum lifter

Element × Reference claim chart

Invention element	EP-33356 07-A4	US-11731 859-B2	EP-36651 09-B1	US-20192 40813-A1	US-20160 53893-A1	US-43858 4-A	US-57721 70-A	EP-39311 43-B1
Cordless handheld vacuum lifter	0.041	.	.	.	.	.	.	.
Vacuum lifter for glass panels	.	.	.	.	.	0.053	.	.
Vacuum lifter for stone panels	.	.	0.055	.	.	.	.	0.053
Flexible sealing lip for vacuum lifter	.	.	.	.	0.049 ¶ p0003	.	.	.
Electric vacuum pump	.	0.04 ¶ p0022	.	.	.	.	0.026 ¶ p0027	.
Vacuum pump runs continuously to maintain grip	.	.	.	0.04 ¶ p0005	.	.	0.032 ¶ p0027	.
Grip on rough surfaces	.	.	.	.	.	.	.	.
Grip on porous surfaces	.	.	.	.	.	.	.	.
Pressure sensor for vacuum lifter	.	.	.	.	.	.	.	.

Invention element	EP-33356 07-A4	US-11731 859-B2	EP-36651 09-B1	US-20192 40813-A1	US-20160 53893-A1	US-43858 4-A	US-57721 70-A	EP-39311 43-B1
Pressure sensor alarms on grip vacuum loss	.	.	.	.	.	.	.	.

Combination / inventive-step analysis

Primary reference: EP-3335607-A4 — discloses: Cordless handheld vacuum lifter

+ **Secondary:** US-11731859-B2 — supplies: Electric vacuum pump

+ **Secondary:** EP-3665109-B1 — supplies: Vacuum lifter for stone panels

+ **Secondary:** US-2019240813-A1 — supplies: Vacuum pump runs continuously to maintain grip

+ **Secondary:** US-2016053893-A1 — supplies: Flexible sealing lip for vacuum lifter

+ **Secondary:** US-438584-A — supplies: Vacuum lifter for glass panels

Elements not disclosed by any reference: Grip on porous surfaces, Grip on rough surfaces, Pressure sensor alarms on grip vacuum loss, Pressure sensor for vacuum lifter

Cited references

1. US-1294103-A — Vacuum lifting device.

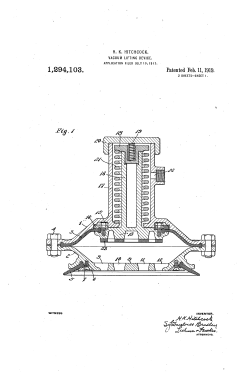
Assignee: PITTSBURGH PLATE GLASS CO **Inventors:** Halbert K Hitchcock

Priority: 1917-07-19 **Filing:** 1917-07-19 **Published:** 1919-02-11 **Family:** 3361658

CPC: B65H3/48, Y10T279/11

Legal status: filed 1917-07-19; priority 1917-07-19; granted 1919-02-11

Prior-art basis: not dated **Match:** 0.779



US-1294103-A · drawing

Relevant passage (whole , similarity 0.779):

Vacuum lifting device.

2. US-1178147-A — Vacuum lifting device.

Assignee: ASBESTOS PROTECTED METAL COMPANY **Inventors:** GARDNER JAMES

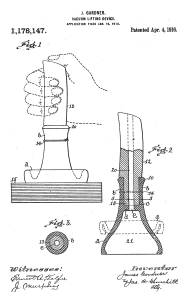
Priority: 1915-01-16 **Filing:** 1915-01-16 **Published:** 1916-04-04 **Family:** 3246136

CPC: B66C1/02, B66C1/02, Y10S206/804, Y10T29/538

Legal status: filed 1915-01-16; priority 1915-01-16; granted 1916-04-04

Prior-art basis: not dated **Match:** 0.779

Reads on: Vacuum lifter for stone panels



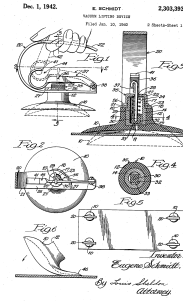
US-1178147-A · drawing

Relevant passage (whole , similarity 0.779):

Vacuum lifting device.

3. US-2303393-A — Vacuum lifting device

Assignee: CHARLES O MOORE **Inventors:** EUGENE SCHMIDT
Priority: 1940-01-10 **Filing:** 1940-01-10 **Published:** 1942-12-01 **Family:** 23215026
CPC: B66C1/0225, B66C1/0212, B66C1/0275, B66C1/0293
Legal status: filed 1940-01-10; priority 1940-01-10; granted 1942-12-01
Prior-art basis: not dated **Match:** 0.77
Reads on: Vacuum lifter for stone panels



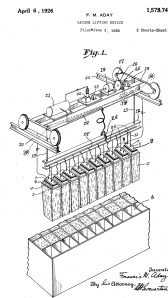
US-2303393-A · drawing

Relevant passage (whole , similarity 0.77):

Vacuum lifting device

4. US-1579744-A — Vacuum lifting device

Assignee: ADAY FRANCIS M **Inventors:** ADAY FRANCIS M
Priority: 1925-06-03 **Filing:** 1925-06-03 **Published:** 1926-04-06 **Family:** 21877324
CPC: B66C1/025, B66C1/0281
Legal status: filed 1925-06-03; priority 1925-06-03; granted 1926-04-06
Prior-art basis: not dated **Match:** 0.77
Reads on: Vacuum lifter for stone panels



US-1579744-A · drawing

Relevant passage (whole , similarity 0.77):

Vacuum lifting device

5. US-762499-A — Vacuum-lifter.

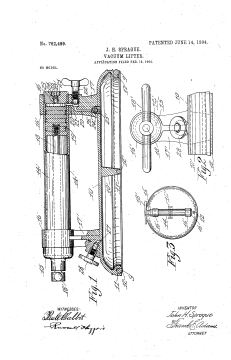
Assignee: SPRAGUE JOHN HOWARD **Inventors:** SPRAGUE JOHN HOWARD

Priority: 1904-02-16 **Filing:** 1904-02-16 **Published:** 1904-06-14 **Family:** 2830985

CPC: B66C1/02

Legal status: filed 1904-02-16; priority 1904-02-16; granted 1904-06-14

Prior-art basis: not dated **Match:** 0.773



US-762499-A · drawing

Relevant passage (whole , similarity 0.773):

Vacuum-lifter.

Appendix — full ranked reference list

#	Publication	Title	Match	Basis	Channels
1	US-1294103-A	Vacuum lifting device.	0.779	not dated	citation, crosslingual, dense, qbe
2	US-1178147-A	Vacuum lifting device.	0.779	not dated	citation, crosslingual, dense, qbe
3	US-2303393-A	Vacuum lifting device	0.77	not dated	citation, crosslingual, dense, qbe
4	US-1579744-A	Vacuum lifting device	0.77	not dated	citation, crosslingual, dense, qbe
5	US-762499-A	Vacuum-lifter.	0.773	not dated	crosslingual, dense, qbe
6	US-2942745-A	Vacuum lift device	0.786	not dated	citation, crosslingual, dense, qbe
7	US-2233350-A	Vacuum lift device	0.786	not dated	citation, crosslingual, dense, qbe
8	US-3055694-A	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
9	US-1691198-A	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
10	US-1705221-A	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
11	US-3227481-A	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
12	US-D953681-S	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
13	US-2010219651-A1	Vacuum lifting device and method of use	0.758	not dated	citation, dense, qbe
14	US-2022153556-A1	Vacuum lifting device	0.767	not dated	citation, crosslingual, dense, qbe
15	US-12115659-B1	Portable vacuum gripper	0.772	not dated	citation, dense, qbe
16	US-11235939-B2	Vacuum lifter	0.738	not dated	citation, crosslingual, dense, qbe
17	US-12122645-B2	Portable vacuum lifter system	0.762	not dated	citation, crosslingual, dense, qbe
18	US-5310232-A	Vacuum lifter	0.759	not dated	citation, crosslingual, dense, qbe

#	Publication	Title	Match	Basis	Channels
19	US-10071885-B2	Vacuum lift attachment	0.755	not dated	citation, cpc, crosslingual, dense, qbe
20	US-12358725-B2	Vacuum lifting device	0.75	not dated	citation, cpc, crosslingual, dense, qbe
21	US-11247858-B2	Vacuum lifter	0.738	not dated	citation, crosslingual, dense, qbe
22	US-10518985-B2	Control systems and methods for vacuum lift equipment	0.786	not dated	citation, cpc, dense, qbe
23	US-3117815-A	Vacuum lifter	0.772	not dated	citation, crosslingual, dense, qbe
24	US-3139300-A	Vacuum-lifting device	0.787	not dated	citation, crosslingual, dense, qbe
25	US-6095581-A	Vacuum lifting device	0.771	not dated	citation, dense, qbe